Analyzing the Influence of Neighbourhood and Other Property Attributes on Calgary’s 2025 Property Assessment Values

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Guiding Question

- **How does assessed value per square foot vary across land use designations and communities?**
- **Which communities show the highest average assessed values and lowest average assessed values?**

Motivation

Our team is passionate about transforming parcel-level data into actionable insights that empower a wide range of stakeholders—analysts, urban planners, real estate professionals, researchers, residents, and even prospective home buyers. By unlocking the value hidden in property data, we aim to support smarter decisions and create a clearer picture of Calgary's housing landscape.

Background

- Property assessments guide municipal tax allocation.
- Influenced by:
    - Parcel size
    - Land-use designation
    - Building characteristics (age, type, floor area etc.)
    - Community/location
- Assessments help analyze socio-economic differences and land-use diversity across communities.
- Prior literature: neighbourhood location is a strong predictor of housing value (Gyourko et al., 2006).
- Parcel size and built form also major contributors.

Data Background

- Data source: City of Calgary – Current Year Property Assessments (2025).
- Population: all taxable parcels across all communities and property types.
- Key variables: Assessed value,Land size,Property type ,Land-use designation and Community

Methods

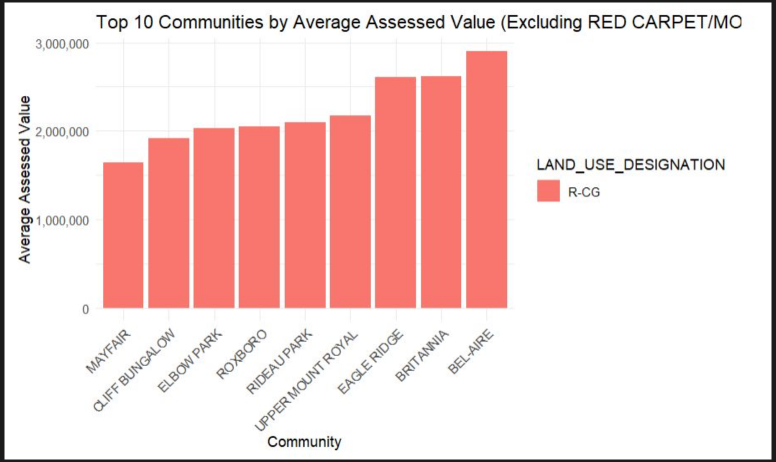
Statistical tools used:

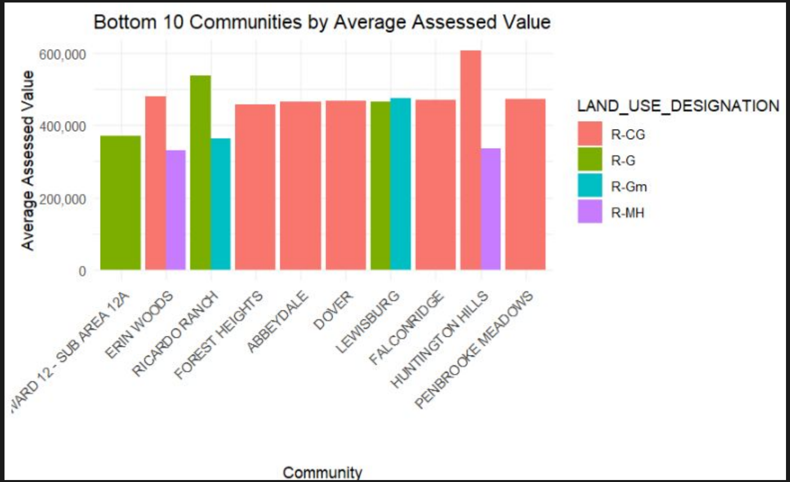
- Descriptive statistics
- Multiple linear regression
- Best subset models
- Diagnostics: residual plots, VIF, assumption checks

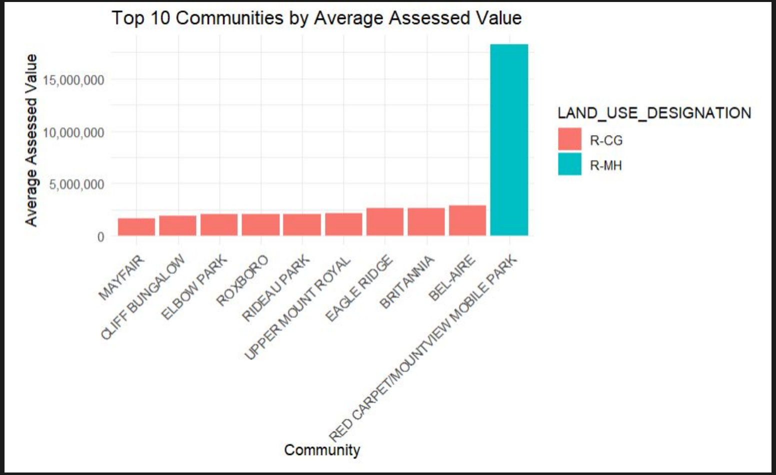
Visualization For Average Assessed Values

Statistical Question: *Which communities show the highest average assessed values and lowest average assessed values?*

Null Hypothesis : No difference in mean assessed values across communities.
Alternate Hypothesis : At least one community differs in mean assessed value.







R-CG is a residential designation that is primarily for rowhouses but also allows for single detached, side-by-side and duplex homes that may include a secondary suite.
The **R-G** district is for a mix of low density housing forms in suburban greenfield locations , including single-detached, side by side, duplex , cottage housing clusters and rowhouse development, all of which may include secondary suite.
R-MH is a residential designation that is for mobile home parks.
The **R-Gm** district is for a range of low density housing forms in suburban greenfield locations , including side by side, duplex , cottage housing clusters and rowhouse development (but excluding single detached), all of which may include a secondary suite.

Visualization For Average Assessed Values

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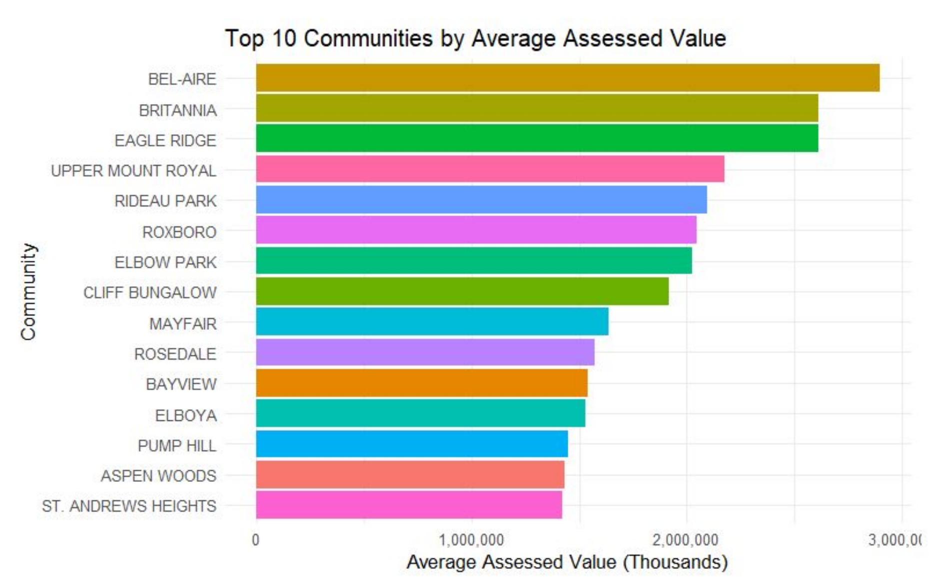
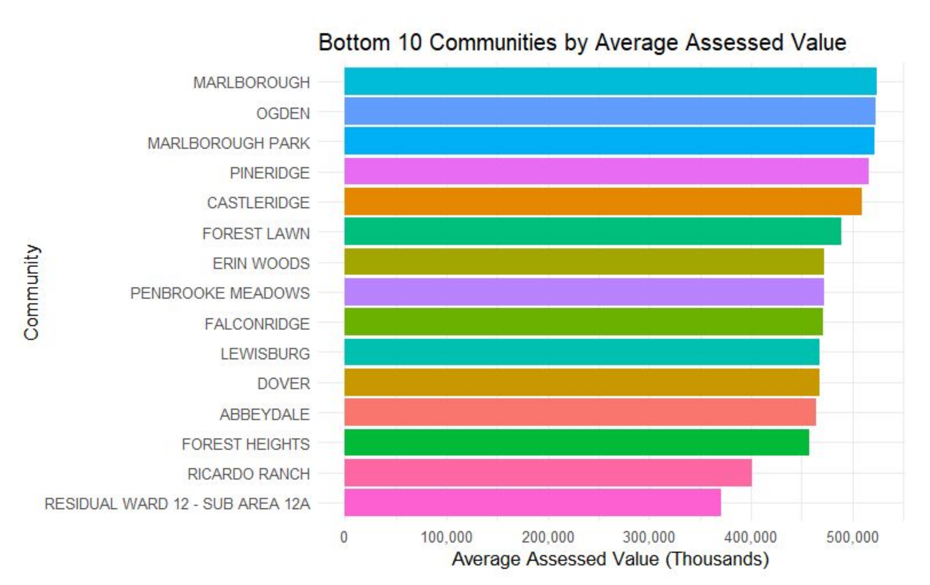


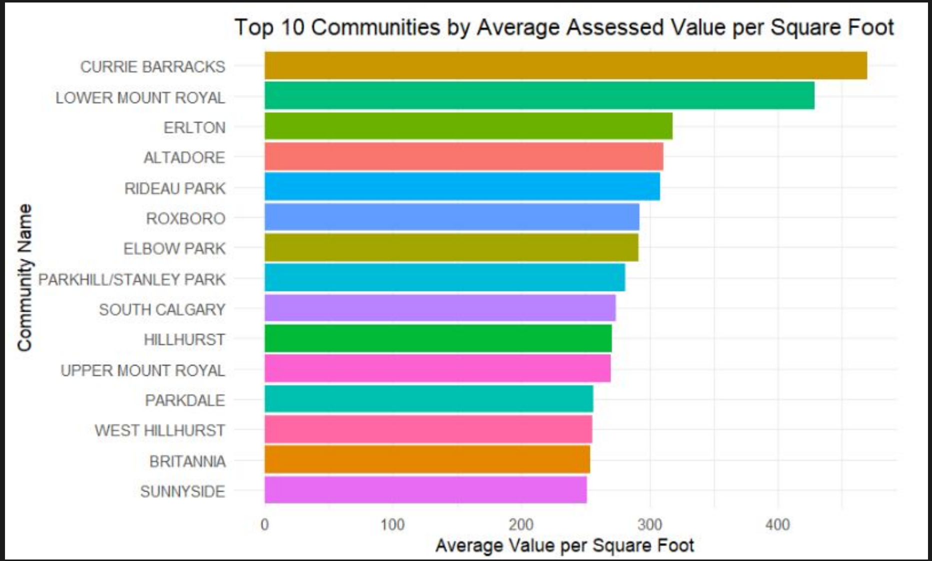


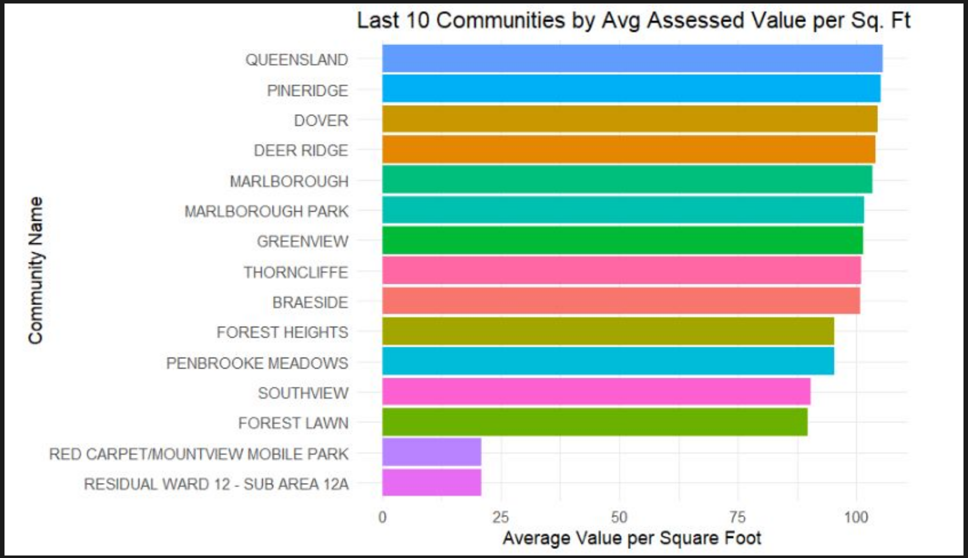
Bar chart for assessed value per square foot across communities

Statistical Question: **How does assessed value per square Foot vary across land use designations and communities?**

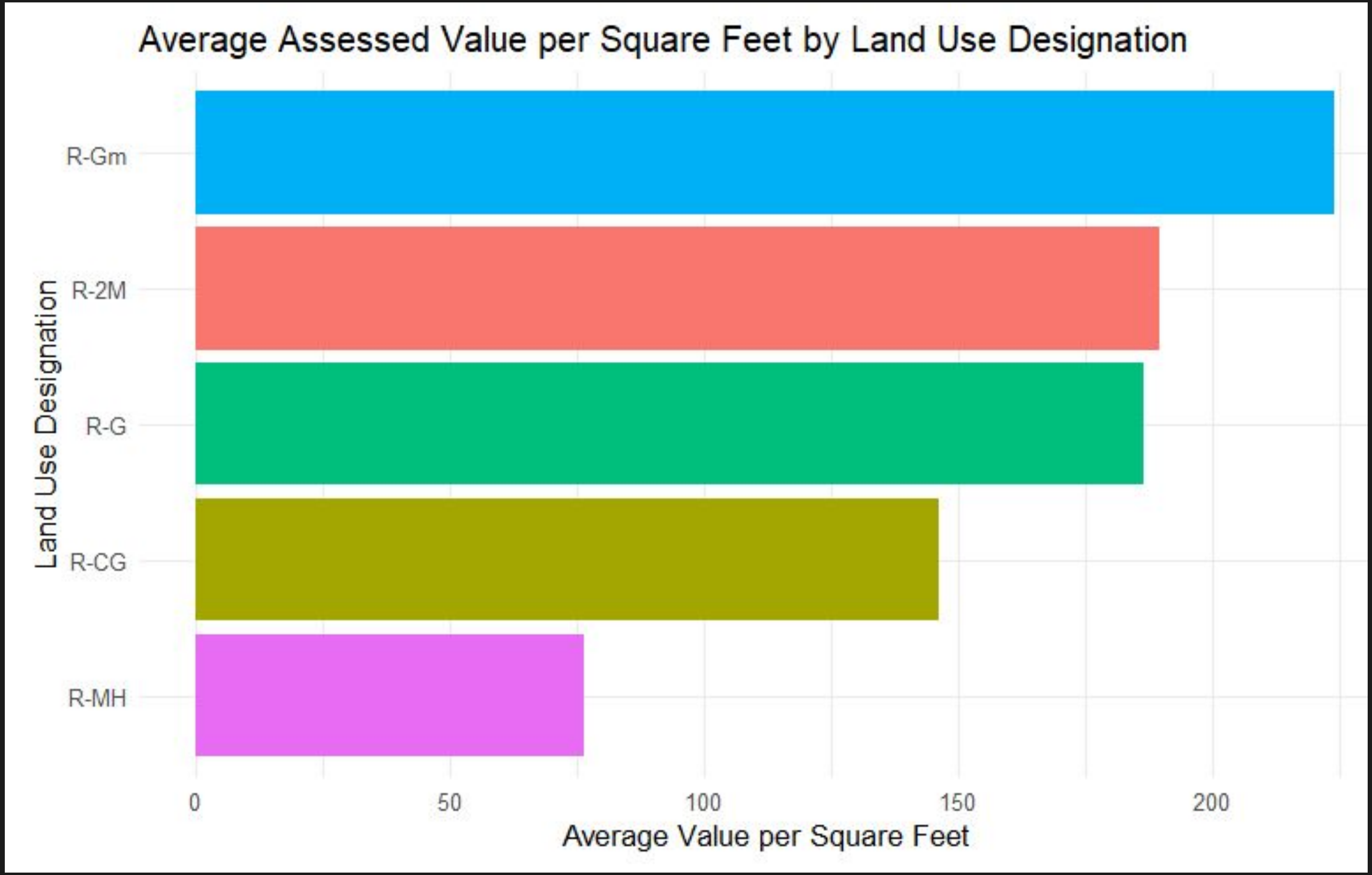
Null Hypothesis : No difference in **assessed value per square foot** across communities.

Alternate Hypothesis : At least one community differs in **assessed value per square foot**.





Bar chart for assessed value per square foot land use designations



Statistical Methods (Regression)

Model:

- Multiple linear regression on:
  - Total assessed value (ASSESSED_VALUE)

Model Features:

- Land Size,Land Use Designation + neighbourhood-level predictors
- Diagnostics: VIF, residuals, assumption checks

Evaluation:

- Adjusted R^2
- Model parsimony via AIC/BIC

Rationale for Regression

- Both outcomes are continuous.
- Allows simultaneous assessment of multiple predictors.
- Controls for confounders such as land use & community location.
- Provides interpretable effect sizes and direction of associations.
- Suitable for assessing valuation drivers and policy relevance.

Multiple Linear Regression Models

full_model <- lm(ASSESSED_VALUE ~ LAND_SIZE_SM + LAND_SIZE_SF + LAND_SIZE_AC + YEAR_OF_CONSTRUCTION + LAND_USE_DESIGNATION + PROPERTY_TYPE + COMM_NAME, data = residential_mut_df)

T test, `LAND_SIZE_AC and PROPERTY_TYPE are not statistically significant with p value of 0.065944 and 0.800345 respectively`, LAND_SIZE_SM is statistically significant but positive to VIF test.

`Adjusted R-squared:  0.5171`

reduced_model <- lm(ASSESSED_VALUE ~ LAND_SIZE_SF + YEAR_OF_CONSTRUCTION + LAND_USE_DESIGNATION + COMM_NAME, data = residential_mut_df)

Most covariates in the model are statistically significant, with the exception of some community-level variables. LAND_SIZE_SM removed from model because of multicollinearity.

`Adjusted R-squared:  0.5566`

reduced_model <- lm(log(ASSESSED_VALUE) ~ LAND_SIZE_SM + LAND_SIZE_SF + YEAR_OF_CONSTRUCTION + LAND_USE_DESIGNATION + COMM_NAME, data = residential_mut_df)

Most covariates in the model are statistically significant, with the exception of some community-level variables.

`Adjusted R-squared:  0.5571`

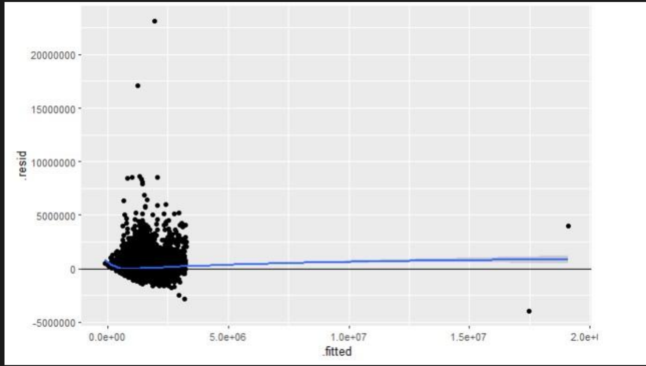
Regression Assumptions

Multicollinearity | Variance Inflation Factor

```
Multicollinearity may be due to LAND SIZE SM LAND SIZE SF
LAND USE DESIGNATIONR-CG LAND USE DESIGNATIONR-G regressors
1 --> COLLINEARITY is detected by the test
```

LAND_SIZE_SM	4.197213e+08	1
LAND_SIZE_SF	4.197227e+08	1
YEAR_OF_CONSTRUCTION	3.346900e+00	0
LAND_USE_DESIGNATIONR-CG	7.208780e+01	1
LAND_USE_DESIGNATIONR-G	1.711400e+01	1
LAND_USE_DESIGNATIONR-Gm	1.978100e+00	0
LAND_USE_DESIGNATIONR-MH	1.270600e+00	0

Linearity | Residual Plot



No evident pattern to suggest non linearity

Multiple Linear Regression Models

Best Subset

```
Length  Class   Mode
     0   NULL   NULL
      rsquare AdjustedR          cp     AIC
[1,] 0.4326350 0.4322911 58110.6422 9169937
[2,] 0.5032824 0.5029798  9815.8439 9126034
[3,] 0.5107621 0.5104581  4710.5228 9121033
[4,] 0.5168522 0.5165506   549.1041 9116899
[5,] 0.5173520 0.5170492   209.4472 9116559
[6,] 0.5173569 0.5170527   208.0640 9116558
[7,] 0.5173570 0.5170513   210.0000 9116560
```

Both AIC and AdjustedR recommends the 6th model

LAND_SIZE_SF YEAR_OF_CONSTRUCTION COMM_NAME

LAND_USE_DESIGNATION

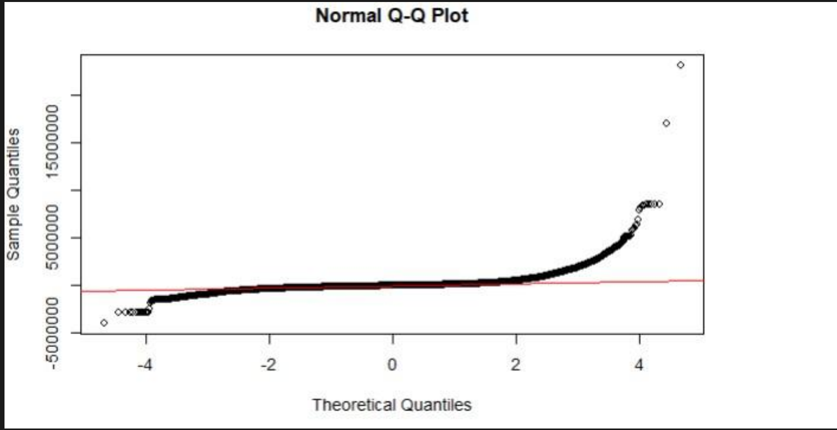
```
Adjusted R-squared:  0.5566
```

Regression Assumptions

```
        Asymptotic one-sample Kolmogorov-Smirnov test

data:  res
D = 0.15945, p-value < 2.2e-16
alternative hypothesis: two-sided
```

Kolmogorov–Smirnov test: Not Normally distributed



As the data appears not normally distributed, we tried log transformation of the assessed value and also robust regression using rlm instead of lm, QQ plot and Kolmogorov test still appear to support that the data is not normally distributed.

Interpretation & Insights

- High mean ≠ wealthy residential area → often large institutional parcels.
- High SD/IQR highlights community heterogeneity and potential valuation disparities.
- Spatial patterns reflect land-use intensity, development mix, and historical zoning.
- Results support need for neighbourhood-specific assessment equity analysis.

Future Directions

- Spatial regression (e.g., GWR, spatial lag/error models).
- Incorporate historical assessment trends.
- Compare assessed values vs. actual market sales.
- Equity analysis: examine over- and under-assessment by neighbourhood demographics.
- Machine-learning models for improved predictive accuracy.
- Factor other covariates like number of amenities in the community, average prices of recently sold properties.

Conclusion

- Parcel characteristics (size, type, land use) significantly shape assessed values.
- Community location remains one of the strongest predictors.
- All of the top 10 communities be average assessed value are established, affluent neighbourhoods in Calgary, concentrated mostly in the southwest quadrant near the Elbow River and Glenmore Reservoir, except Rosedale, which is in the northwest quadrant.
- Most of the bottom 10 communities are in southeast Calgary (Forest Lawn, Erin Woods, Penbrooke Meadows, Dover, Forest Heights, Ricardo Ranch).A few are in the northeast quadrant (Falconridge, Abbeydale, Lewisburg).
- Ricardo Ranch and Residual Ward 12 Sub Area 12A are new or planned developments in the city's southeast growth corridor.

References

City of Calgary Open Data Portal: Current Year Property Assessments (Parcel). Current Year Property Assessments (Parcel) | Open Calgary

Gyourko, J., Mayer, C., & Sinai, T. (2006). Superstar Cities(NBER Working Paper No. 12355). National Bureau of Economic Research. Microsoft Word - superstar_cities_06-16-06-final.doc

Thank you